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The Foundations: Logic, Proofs, Sets and Logic Function



Learning Objectives

At the end of the lesson, the students must be able to:

- Define key terms in logic, including premise, conclusion, and argument.
- Distinguish between arguments and non-arguments in written texts.
- Apply logical reasoning skills to real-world issues and debates, demonstrating the relevance of logic in everyday decision-making.



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What makes human special?

- we are RATIONAL



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What does rationality consist in?

REASONING

CR
CRITICISMS

- making claims and backing them up with reasons, acting in accord with reasons and beliefs, drawing inferences from available evidence, and so on.



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What is logic?

According to Matthew Knachel, logic is the discipline that aims to distinguish good reasoning from bad.



Example

Let's consider Hitler. He persuaded an entire nation to go along with a variety of proposals that were not only false but downright evil. You won't be surprised to hear that if you examine it critically, his reasoning does not pass logical muster. Hitler's arguments were effective, but not logically correct. Moreover, his persuasive techniques go beyond reasoning in the sense of backing up claims with reasons. Hitler relied on threats, emotional manipulation, unsupported assertions, etc. There are many rhetorical tricks one can use to persuade.



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BASIC NOTIONS ABOUT REASONING

* Propositions

- Things we claim, state, assert.
- Kinds of things that can be true or false.
- They are expressed by declarative sentences



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*Argument S

- a set of propositions, one of which, the conclusion, is (supposed to be) supported by the others, the premises.



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ANALYSIS OF ARGUMENTS

You shouldn't eat at McDonald's. Why? First of all, because they pay their workers very low wages. Second, the animals that provide their meat are raised in deplorable conditions. Finally, the food is extremely unhealthy.



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Explicate Arguments

1. McDonald's pays their workers very low wages.
2. The animals that provide McDonald's meat are raised in deplorable conditions.
3. McDonald's food is very unhealthy.

/∴ You shouldn't eat at McDonald's.



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Premises Marker

- a signal word or phrase used in logical arguments to indicate that what follows is a premises.
- because, since, for, given that, as and etc.



Conclusion Marker

- a word or phrase used in logical arguments to signal that what follows is the conclusion.
- consequently, thus, it follows that, which implies that, “therefore”



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ANALYSIS OF ARGUMENTS

The universe is vast and complex. And yet does it not also display an astonishing degree of order? The planets orbit the sun according to regular laws, and animals' minutest parts are arranged precisely to serve their purposes. Such order and complexity cannot arise at random. The universe must therefore be the product of a Designer of enormous power and intellect, whom we call God.



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Explicate Arguments

1. The universe is vast and complex.
 2. The universe displays an astonishing degree of order.
 3. The planets orbit the sun according to regular laws.
 4. Animals' minutest parts are arranged precisely to serve their purposes.
 5. Such order and complexity cannot arise at random.
- $\therefore$ The universe must be the product of a designer of enormous power and intellect: God.



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ANALYSIS OF ARGUMENTS

She must be a good student because
she always gets to class on time.



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Explicate Arguments

1. Students who arrive on time are responsible and good students."

$\therefore$ She is a good student.



Tacit Premices

- unstated or implied assumptions in an argument that are not explicitly mentioned but are necessary for the argument to make sense. These premises are assumed to be understood or accepted by both the speaker and the audience without needing to be directly expressed.



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Kinds of Premises

- Independent premises
- Intermediate Premises
- Joint Premises



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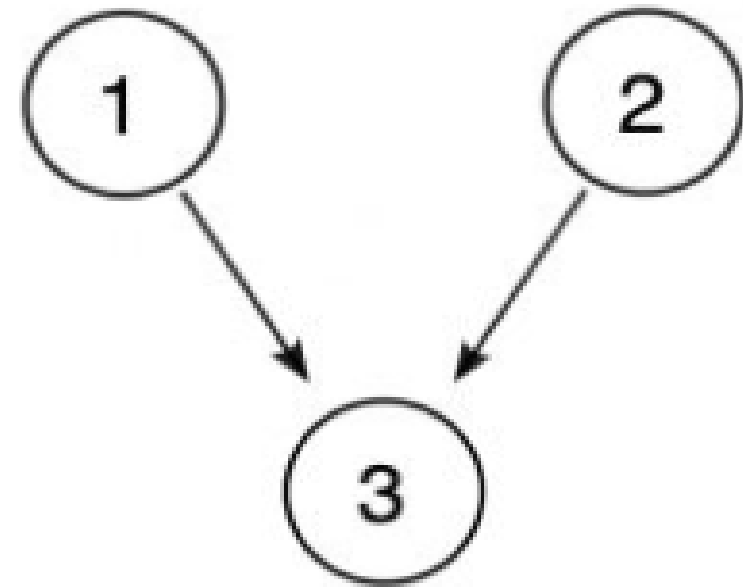
Independent Premises

statements in an argument that support the conclusion on their own, without relying on one another.



Example

- ① America's invasion of Iraq was an act of aggression, not self-defense. In addition, ② it was unreasonable to expect that the benefits of the war would outweigh the inevitable horrors it would unleash. Therefore, ③ the Iraq War was not a just war.





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Intermediate Premises

statements in an argument that serve as links between the main premises and the conclusion.



Example

- ① Poets are mere “imitators” whose works obscure the truth; hence, ② they have a corrupting influence on the souls of citizens. ③ Poets should therefore be banned from the ideal city-state.





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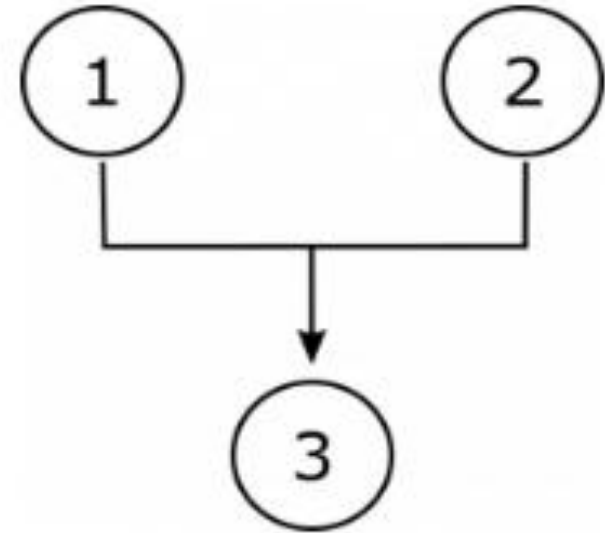
Joint Premises

Two or more premises in an argument that work together to support a single conclusion.



Example

- ① If true artificial intelligence is possible, then one must be able to program a computer to be conscious. ② But it's impossible to program consciousness. Therefore, ③ true artificial intelligence is impossible.





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REFERENCE:

<https://press.rebus.community/intro-to-phil-logic/chapter/chapter-1/>